Advanced Statistical Methods II

Lecture III: Conditional Imputation and Uncertainty

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Outline of Today's Lecture

Recap and motivation

From marginal thinking to conditional thinking

1. Conditional mean imputation

2. Regression-based imputation

3. Stochastic imputation

Why single imputation underestimates uncertainty

4. Full conditional distributions

Imputation as prediction, not truth recovery

Key Takeaway Messages

Where we are in the course

Missingness can affect inference, the validity of complete-case analysis, and the likelihood we should use.

Today we ask a more constructive question:

If some values are missing, can we fill in plausible values?

The word plausible is crucial.

- Not every filled-in value is statistically meaningful.
- A good imputation should use information in the observed data.
- It should also acknowledge that the missing value is uncertain.

See Little and Rubin (2019), Chs. 1–4; van Buuren (2018), Chs. 1–2.

A small motivating example

Suppose a clinical study records:

Patient	Age	Baseline severity	6-month outcome
1	43	Low	12
2	67	High	?
3	51	Medium	19
4	70	High	?
5	38	Low	10

What should replace the two question marks?

Possible answers:

- the overall mean of observed outcomes;
- the mean among patients with similar severity;
- a predicted value from a regression model;
- a random draw from a predictive distribution.

Focus of Today's Lecture

Imputation means replacing missing values by one or more plausible values, constructed using a statistical methodology/philosophy.

Today we focus on four ideas:

1. conditional mean imputation;
2. regression-based imputation;
3. stochastic imputation;
4. full conditional distributions and the beginning of iterative imputation.

Recap: notations

Let the full data be

$$Y = (Y_{\mathrm{obs}}, Y_{\mathrm{mis}}),$$

where:

- Y_{obs} denotes the actually observed part;
- Y_{mis} denotes the missing part;
- R denotes the response/missingness indicator;
- θ denotes parameters of the data model.

The central conditional distribution is

$$p(Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}}, R, \theta).$$

Under *ignorable missingness assumptions*, we work with

$$p(Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}}, \theta).$$

A bad default: unconditional mean imputation

A common first attempt is:

$$\tilde{Y}_i = \bar{Y}_{\mathrm{obs}} \quad \text{whenever } Y_i \text{ is missing.}$$

It uses only the marginal distribution of the observed values of Y.

Disadvantages:

- ignores predictors that may be related to Y;
- can distort correlations and regression slopes;
- treats imputed values as if they were known exactly;
- artificially reduces variation.

See Little and Rubin (2019), Ch. 4; Enders (2010), Ch. 3.

Conditional thinking

Instead of asking:

What is a typical value of Y?

we ask:

What is a plausible value of Y given what we observed?

That is,

$$p(Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}}, \theta).$$

Imputation should preserve relationships among variables as much as possible.

For example:

- income may depend on age, education, and occupation;
- health outcome may depend on baseline severity;
- exam score may depend on attendance and previous scores.

Suppose X is fully observed and Y has missing values.

If X and Y are positively associated, then two people with different X values should not receive the same imputed Y value.

Case	Observed X	Reasonable imputed Y?
A	Low X	Lower plausible range
B	Medium X	Middle plausible range
C	High X	Higher plausible range

The conditional distribution contains two pieces

For a missing scalar value Y_i, conditional on observed information X_i, think of

$$Y_i \mid X_i = x_i.$$

This distribution has:

1. a conditional center:
$$\mathbb{E}(Y_i \mid X_i = x_i),$$
2. a conditional spread:
$$\mathbb{V}\mathrm{ar}(Y_i \mid X_i = x_i).$$

Many simple imputation methods use the conditional center but ignore the conditional spread.

1. Conditional mean imputation

The most direct conditional imputation rule is

$$\tilde{Y}_{\mathrm{mis}} = \mathbb{E}\left(Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}}, \widehat{\theta}\right).$$

Replace each missing value by the conditional expectation under a fitted model.

This is better than unconditional mean imputation when the conditioning variables carry information.

But it is still deterministic:

$$\text{same observed information} \;\Rightarrow\; \text{same imputed value.}$$

Conditional mean imputation still can not capture the uncertainty fully.

Example: Bivariate Normal model

Suppose

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix} \right).$$

Then

$$Y \mid X = x \sim N\left(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1 - \rho^2) \right).$$

Conditional mean imputation uses

$$\tilde{Y} = \widehat{\mu}_Y + \widehat{\rho}\frac{\widehat{\sigma}_Y}{\widehat{\sigma}_X}(x - \widehat{\mu}_X).$$

Standard multivariate normal conditioning; see Schafer (1997), Ch. 5.

A few remarks.

- If $\rho = 0$, then X gives no information about Y.
- If $|\rho|$ is large, then X gives strong information about Y.
- The conditional variance is

$$\sigma_Y^2(1 - \rho^2).$$

Even when $|\rho|$ is large, the conditional variance is usually not zero. A missing value remains uncertain.

A basic calculation

Suppose the fitted conditional distribution is

$$Y \mid X = x \sim N(10 + 2x, 4).$$

For a missing Y with $X = 3$:

$$\mathbb{E}(Y \mid X = 3) = 10 + 2(3) = 16.$$

So conditional mean imputation gives

$$\tilde{Y} = 16.$$

But the conditional distribution is

$$Y \mid X = 3 \sim N(16, 4).$$

The imputed value 16 is only the center of plausible values. Values like 13.8, 15.2, 17.5, or 19.1 are also plausible.

Strengths	Weaknesses
Uses observed information	Understates variability
Simple to explain	Treats predictions as observed truth
Often improves over mean imputation	Can distort standard errors
Related to regression ideas	Does not propagate parameter uncertainty

> It may be useful for prediction of missing entries, but it is generally not enough for valid statistical inference.

2. Regression imputation: the basic model

Suppose Y has missing values and X is fully observed.

Fit the model using complete cases:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2).$$

For a missing Y_i, impute

$$\tilde{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i.$$

This is just conditional mean imputation under a linear regression model.

See Little and Rubin (2019), Ch. 4; van Buuren (2018), Secs. 1.3–1.4.

Regression imputation with several predictors

More generally,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \varepsilon_i.$$

Then

$$\tilde{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_{i1} + \cdots + \widehat{\beta}_p X_{ip}.$$

Useful predictors might include:

- variables related to the missing variable;
- variables related to the missingness mechanism;
- design variables, group indicators, or baseline measurements.

Good imputation models should be at least as rich as the analysis model, and often richer.

This principle is emphasized in Rubin (1987), Schafer (1997), and Carpenter and Kenward (2013).

Why include variables related to missingness?

Recall MAR from Lecture II:

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}) = \mathbb{P}(R \mid Y_{\text{obs}}).$$

If missingness depends on a fully observed variable X, then conditioning on X can make the missingness more manageable.

Example

High-severity patients are more likely to miss follow-up, and baseline severity is observed.

Then an imputation model for the follow-up outcome should include baseline severity.

Ignoring such variables can turn an apparently harmless imputation into a biased analysis.

Regression imputation is model-based

Regression imputation depends on assumptions:

- the regression form is approximately correct;
- the residual distribution is reasonable;
- the fitted relationship for observed cases applies to missing cases;
- relevant predictors are included;

A common mistake

After deterministic regression imputation, one might analyze the completed dataset as if nothing were missing.

For example:

$$\text{Fit } Y \sim X \text{ after replacing missing } Y \text{ by } \widehat{Y}.$$

> This is usually invalid for inference because the imputed values have no residual noise and no parameter uncertainty.

Consequences:

- standard errors may be too small;
- confidence intervals may be too narrow;
- fitted associations can be exaggerated or distorted.

See Little and Rubin (2019), Ch. 4; Sterne et al. (2009).

3. Stochastic Imputation

A simple way to respect conditional spread is stochastic regression imputation.

Fit

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2).$$

For missing Y_i, draw

$$\varepsilon_i^* \sim N(0, \widehat{\sigma}^2), \quad \text{(adding residual noise)}$$

and impute

$$\tilde{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i + \varepsilon_i^*.$$

Now imputed values have realistic scatter around the regression line.

Conditional mean vs stochastic imputation

Conditional mean imputation	Stochastic imputation
Uses $\mathbb{E}(Y \mid X)$	Draws from $Y \mid X$
No residual noise	Adds residual noise
Creates too-smooth completed data	Preserves scatter better
Good for simple prediction	Better for representing uncertainty in missing values
Still ignores parameter uncertainty	May still ignore parameter uncertainty unless parameters are also drawn

> Stochastic imputation is an improvement, but one randomly completed dataset is still not the whole uncertainty story.

Predictive distribution viewpoint

The most principled target is not simply

$$\mathbb{E}(Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}}, \widehat{\theta}).$$

It is the predictive distribution

$$p(Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}}).$$

If θ is unknown, then formally

$$p(Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}}) = \int p(Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}}, \theta)\, p(\theta \mid Y_{\mathrm{obs}})\, d\theta.$$

Uncertainty comes from two sources: residual uncertainty about missing values and parameter uncertainty about the model.

This predictive-distribution view is central in Rubin (1987) and Schafer (1997).

Two sources of uncertainty

Source	Meaning
Residual uncertainty	Even if the model parameters were known, a missing value is not exactly predictable.
Parameter uncertainty	We estimate the model parameters from incomplete data, so $\widehat{\theta}$ itself is uncertain.

Conditional mean imputation ignores both. Simple stochastic imputation handles residual uncertainty but usually not parameter uncertainty.

This motivates multiple imputation.

Mini example: why one imputation is unstable

Suppose

$$Y \mid X = 3 \sim N(16, 4).$$

A stochastic imputation might draw

$$\tilde{Y} = 15.1.$$

Another equally valid draw might be

$$\tilde{Y} = 18.4.$$

A third might be

$$\tilde{Y} = 13.6.$$

If our final conclusion changes drastically depending on one random draw, we should not base inference on only one completed dataset.

The completed-data illusion

After imputation, the dataset visually appears complete.

ID	X	Y	Missing?	Status
1	2.1	13.4	No	observed
2	3.0	16.0	Yes	imputed
3	4.4	21.2	No	observed
4	1.8	14.3	Yes	imputed

But the two imputed values are not the same kind of information as the observed values.

Why variance is underestimated: intuition

If we replace a missing value by a single number, then downstream analysis treats that number as fixed.

But before imputation, it should be a distribution:

$$Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}}.$$

Single imputation collapses that distribution to a point:

$$Y_{\mathrm{mis}} \mid Y_{\mathrm{obs}} \quad \longrightarrow \quad \tilde{Y}_{\mathrm{mis}}.$$

A preview of multiple imputation

Multiple imputation creates several completed datasets:

$$Y^{(1)}, Y^{(2)}, \ldots, Y^{(m)}.$$

Each has different plausible values for Y_{mis}.

Analyze each dataset, then combine the results.

Imputation	Estimate	Standard error
1	$\widehat{Q}^{(1)}$	$U^{(1)}$
2	$\widehat{Q}^{(2)}$	$U^{(2)}$
$\vdots$	$\vdots$	$\vdots$
m	$\widehat{Q}^{(m)}$	$U^{(m)}$

Multiple imputation was formalized by Rubin (1987); see also van Buuren (2018).

Rubin's combining rules: only a preview

Let $\widehat{Q}^{(j)}$ be the estimate from imputed dataset j.

The combined estimate is

$$\bar{Q} = \frac{1}{m} \sum_{j=1}^{m} \widehat{Q}^{(j)}.$$

The total variance combines:

- within-imputation variance;
- between-imputation variance.

> We will not develop multiple imputation fully today, but this formula shows why several plausible datasets are useful.

When several variables have missing values

So far, we imagined one variable Y with missing values and predictors X fully observed.

But in real data, several variables may be incomplete:

$$Y_1, Y_2, \ldots, Y_p.$$

Example:

ID	Age	Income	Health score	Follow-up
1	43	52	78	82
2	67	?	61	?
3	?	44	70	73
4	70	69	?	?

Now there is no single obvious response variable and no single obvious predictor set.

4. Full conditional distributions

A full conditional distribution describes one variable given all the others.

For variable Y_j:

$$p(Y_j \mid Y_{-j}, \theta_j),$$

where Y_{-j} denotes all variables except Y_j.

Examples:

$$\begin{aligned} &Y_1 \mid Y_2, Y_3, \ldots, Y_p, \\ &Y_2 \mid Y_1, Y_3, \ldots, Y_p, \\ &\quad\vdots \\ &Y_p \mid Y_1, Y_2, \ldots, Y_{p-1}. \end{aligned}$$

The full conditional idea lets us build imputation models variable by variable.

Different variables need different models

Full conditional imputation is flexible because each variable can use a suitable model.

Variable type	Possible conditional model
Continuous	Linear regression
Binary	Logistic regression
Categorical	Multinomial regression
Count	Poisson or negative binomial model
Skewed continuous	Predictive mean matching or transformations

See van Buuren (2018), Chs. 3–4; van Buuren and Groothuis-Oudshoorn (2011).

The chained-equations idea

A common practical strategy is:

1. initialize missing values crudely;
2. impute Y_1 given all other variables;
3. impute Y_2 given all other variables;
4. continue through Y_p;
5. repeat the cycle several times.

This is often called fully conditional specification or multivariate imputation by chained equations.

Each step is simple. The overall algorithm can handle complex missingness patterns.

See van Buuren (2018); implementation in R is described by van Buuren and Groothuis-Oudshoorn (2011).

A conceptual algorithm

One cycle of chained imputation

1. Regress income on age, health score, and follow-up.
2. Draw plausible missing income values.
3. Regress health score on age, income, and follow-up.
4. Draw plausible missing health scores.
5. Regress follow-up on age, income, and health score.
6. Draw plausible missing follow-up values.

After several cycles, we retain one completed dataset.

Repeat the whole procedure m times to obtain multiple imputed datasets.

Why the word “conditional” matters again

Full conditional imputation says:

Use everything currently known to impute one variable at a time.

This protects important relationships such as:

- association between income and education;
- association between baseline and follow-up measurements;
- association between risk score and disease outcome;
- group differences due to treatment, region, or cohort.

> If a scientifically important relationship is absent from the imputation model, the completed data may weaken or erase it.

Scientific caution

Imputation is most credible when:

- missingness is plausibly MCAR or MAR after conditioning on observed variables;
- the imputation model includes important predictors;
- the imputation model respects the form of the data;
- sensitivity analyses are considered when MNAR is plausible;
- uncertainty due to missingness is reported honestly.

> In public health, clinical trials, survey science, and policy evaluation, overconfident imputation can become overconfident decision-making.

See National Research Council (2010) and Sterne et al. (2009) for applied cautions.

Connecting to the EM algorithm

The next major topic is the EM algorithm.

It also begins with missing or latent quantities.

But instead of filling in a single completed dataset, EM repeatedly computes conditional expectations of missing-data quantities.

Imputation viewpoint	EM viewpoint
Fill in plausible missing values	Average over missing values
Work with completed datasets	Work with expected complete-data log-likelihood
Represents uncertainty through repeated draws	Uses conditional expectation to optimize likelihood

Both ideas are built on $Y_{\text{mis}} \mid Y_{\text{obs}}, \theta$.

A few examples

For each statement, decide whether it is reasonable or flawed.

1. "I imputed missing incomes by the overall mean, then used the completed data for a regression."
2. "I imputed missing follow-up outcomes using baseline severity, age, and treatment group."
3. "I created one stochastic imputed dataset and reported ordinary standard errors."
4. "I used several imputed datasets and combined estimates to reflect between-imputation variation."

> Which choices respect conditional information? Which choices respect uncertainty?

Answers

1. Flawed: overall mean imputation ignores conditional structure and understates variability.
2. Reasonable: the imputation uses observed predictors that are scientifically relevant to the missing outcome.
3. Flawed for inference: stochastic values add residual scatter, but one completed dataset does not fully represent imputation uncertainty.
4. Reasonable: multiple completed datasets can represent missing-data uncertainty when the imputation model is appropriate.

> Good imputation needs both conditional information and honest uncertainty accounting.

Key Takeaway Messages

1. Imputation should be conditional on observed information, not merely based on marginal averages.
2. Conditional mean imputation uses the center of $Y_{\text{mis}} \mid Y_{\text{obs}}, \widehat{\theta}$.
3. Regression imputation is conditional mean imputation under a regression model.
4. Stochastic imputation draws from a predictive distribution and better preserves scatter.
5. Single imputation usually underestimates uncertainty.
6. Full conditional distributions allow variable-by-variable imputation in multivariate data.

Suggested reading after Lecture III

For this lecture, the most relevant readings are:

- Little and Rubin (2019), Chapters 1–4: core missing-data framework and simple methods.
- van Buuren (2018), Chapters 1–4: practical imputation, full conditional specification, and chained equations.
- van Buuren and Groothuis-Oudshoorn (2011): R implementation of chained equations.

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